

AdJustMoment: Customize Your Ad Watching Experience

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ABSTRACT

Ads are ubiquitous on online video platforms, providing significant commercial value but also often interrupting viewers at inopportune moments. This can be particularly disruptive for mobile users, whose diverse contexts of use can mean time for video watching is limited. Prior research indicates that allowing users to negotiate or defer interruptions can reduce disruption. Inspired by this, we developed AdJustMoment, a mobile video player application that imports content from online platforms and offers users the option to defer ads through "Snooze" or "Ads-debt" - user-determined ad-viewing times. Our preliminary qualitative feedback indicates a preference for "Ads-debt", as users felt it provided greater control over when ads appear. Additionally, users also desire to receive positive feedback upon completing an ad, as it serves as a motivational factor.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI); Ubiquitous and mobile computing design and evaluation methods.**

KEYWORDS

Ads; Mobile HCI; Interruptions; Deferral

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1 INTRODUCTION

The proliferation of mobile devices has resulted in a significant increase in mobile entertainment consumption, including video watching. Research has shown that mobile video consumption

often takes place during periods of waiting [4], time-killing [2], commuting [5], or simply for a mental break [1]. However, the anytime, anywhere nature of mobile video consumption also means that viewing time is often fragmented and limited. This fragmentation is exacerbated by interruptive video ads which not only curtail the available viewing time but, if poorly timed, can disrupt the immersive viewing experience. This not only degrades the overall video-watching experience but may also undermine the primary purpose of ads, which is to attract user attention. Considering that only a small proportion of mobile users subscribe to ad-free video plans [3] [7], with nearly 97% of the users have to contend with the ad interruptions, poorly timed ads can lead to less satisfying viewing experiences for the majority of mobile video audiences. Therefore, we propose a user-centric approach to ad timing, aiming to reduce viewer disturbance and potentially enhance their willingness to engage with ads.

Past studies on interruptibility and mobile notification management have demonstrated that allowing interruptions to be deferred or snoozed can significantly enhance user experience. For instance, experiments where individuals could negotiate interruption timings with their devices yielded better task performance and user preference outcomes [6]. Similarly, when participants were given control over notification snooze durations, positive results were observed [9]. Weber et al. [8]'s research on notification deferral found snoozing to be the most popular method among users for managing interruption postponement. Inspired by these findings, we hypothesized that enabling users to snooze or choose the time for ad viewing could safeguard the integrity of the video watching experience and allow users to devote more cognitive resources to ad viewing when prepared to do so. However, the feasibility and user acceptance of snoozing or timing negotiation for ads have not been explored in prior studies, leaving these aspects uncertain. Therefore, our research aims to investigate the concept of snoozing and timing negotiation for ads in mobile video playback, and assess whether these methods can improve overall viewing satisfaction and potentially enhance user engagement with ads. In essence, we explore whether turning intrusive ad insertions into user-scheduled occurrences can benefit the user's viewing experience.

In order to carry out this investigation, we developed an mobile video playback application named AdJustMoment. This application allows users to control the timing of ad appearances while watching videos on mobile devices. The app incorporates two key mechanisms: 1) Snooze, which enables users to delay ads to a later time, and 2) Ads-debt, a feature that allows users to "accumulate" ads

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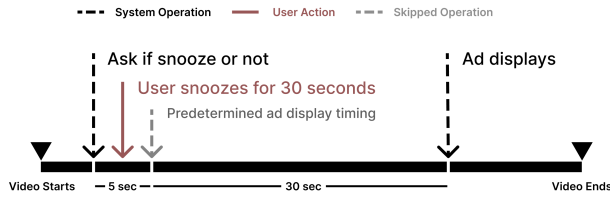


Figure 1: The video playback timeline in Snooze mode (taking a 30-second snooze as an example).

and decide when to watch them after completing several videos. To understand users' experiences with these mechanisms and gauge their feasibility, we solicited informal feedback from mobile video platform users. Our preliminary findings indicate a preference for the Ads-debt feature as it offered users a higher level of control over ad timing. Further findings include that users who use the video playback applications wish to have a rewarding mechanism when they watch an ad, or else it would create a feeling of loss.

The expected contributions of a full study are the assessment of perceived feasibility and the user experience regarding two strategies for ad deferral in mobile video platforms. Through our future full study, we aim to determine whether these widely adopted deferral strategies can be successfully applied to mobile video ads, and if they can enhance the mobile video viewing experience while simultaneously facilitating user engagement with ads.

2 METHODOLOGY

We developed a mobile application named AdJustMoment, which is a video playback software that allows users to independently determine the timing of ad viewing. The two ads video deferral methods developed in this study are inspired by previous research conducted by scholars who have designed or proposed mechanisms for deferring notifications. In the application, we have created a playlist comprised of pre-existing videos from the YouTube platform. As users engage with this playlist via AdJustMoment, they will receive alerts signifying upcoming ads at specific moments. During these moments, users can exercise the deferral features, namely "Snooze" and "Ads-debt," to manage the timing of the ads. Figure 3 illustrates the sequence of user interactions with AdJustMoment, starting from the selection of the initial video to the utilization of "Snooze" or "Ads-debt" features to defer ads.

To evaluate the two deferral methods, we mimic the ad timing structure of YouTube, inserting ads collected from the platform itself. Ad videos are played by switching the video source at designated moments. When encountering ads during video playback, users employ the deferral methods they previously selected to decide when they will watch the ad videos.

2.1 Snooze

The Snooze feature's operational flow and user interaction during video playback is depicted in Figure 1. (See C in figure 3) As a user watches a video, a "Snooze" prompt appears five seconds prior to

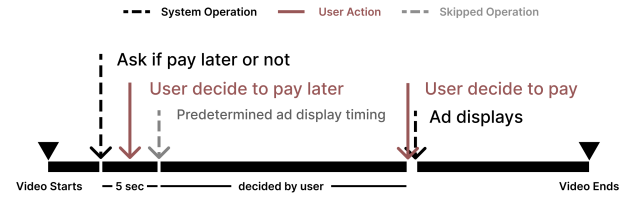


Figure 2: The video playback timeline in Ads-debt mode.

the pre-determined ad insertion time. On clicking this button, the video pauses, and a duration selection spinner for snoozing appears. Snooze durations are available in thirty-second increments, which grant users more control. However, the delay cannot exceed the duration of the currently playing video, meaning users must finish watching the ad before the video concludes. If the user chooses not to snooze, the video continues to play, and the ad will display at the predetermined time.

2.2 Ads-debt

The "Ads-debt" concept is modeled on the metaphor of financial debt. This means that the user can choose to delay viewing the ad when it initially appears. In doing so, the ad is added to the user's "ads-debt" list, which must be paid off within a certain timeframe, say, within 24 hours. In this method, ads are conceptualized as a form of debt that can be deferred within certain limits but must eventually be repaid after a specific number of deferrals.

The flow of video playback and user interaction in the "Ads-debt" mode is illustrated in Figure 2. When a user selects the "Ads-debt" option, a prompt with the text "Save in my ads debt" appears five seconds before the default ad display time, inquiring if the user wants to defer viewing that particular ad. If the user does not click this button, the video continues to play and the ad will be shown at the initially scheduled time. However, if the user chooses to add the ad to their debt, the system places it in a queue, providing the user with flexibility in deciding when to view the accumulated pending ads.

This queue has a specific capacity for ads. Once the queue is full, the user will be unable to defer any more ads and must watch at least one ad to free up space for additional deferrals. When the "Ads-debt" mode is activated, a circular button appears in the bottom-right corner of the interface, with the number on the button indicating the current count of deferred ads. (See D in figure 3) If the user wishes to start repaying their "ads-debt", they can simply tap this button, triggering the display of the deferred ads. Tapping the button again while an ad is playing indicates that the user wants to return to the original video once the current ad has finished, thus allowing the user to continue viewing their chosen content.

2.3 Interface

The interface of the system is shown in Figure 3, which depicts examples of using both the "Snooze" (C) and "Ads-debt" (D) features respectively. (Figure 3 A) When users engage with this application,

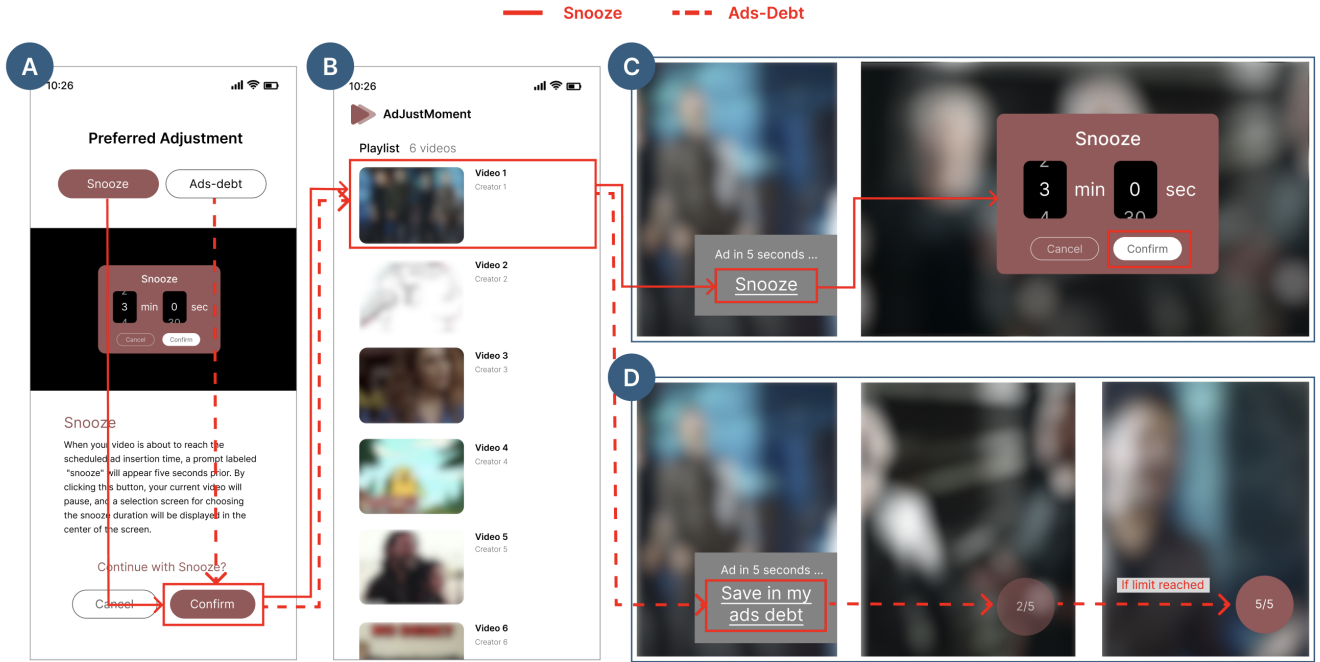


Figure 3: The operational flow under different modes of AdjustMoment.

they are initially guided to select an ad deferral method. Accompanying the preview, there will be a concise text explanation that offers an overview of the deferral method. If users are content with their selection, they can confirm their choice by clicking the confirmation button. (Figure 3 B) Once users have selected their preferred deferral method, they navigate to the video selection page, where they encounter a pre-curated playlist of videos sourced from the YouTube platform. Users can then select the initial video they wish to view from this playlist.

3 PRELIMINARY EVALUATIONS

3.1 Evaluation

In our preliminary evaluation, we obtained informal feedback from 9 students studying in the Department of Computer Science at National Yang Ming Chiao Tung University. On average, they spend 3 hours per day watching mobile videos containing ads. We introduced the two deferral methods to these users and sought their opinion regarding the feasibility of these methods, any considerations they may have, and their overall impressions. All participants interacted with the system using both deferral methods.

Following this trial period, we solicited informal feedback from the users on the perceived feasibility of these mechanisms. The general sentiment from participants was positive, with users expressing approval of the concept and finding value in the greater flexibility these deferral methods provide in determining when to watch ads. This added control over the ad-viewing experience was seen to reduce the disruption usually experienced during video viewing.

For example, P2 commented, "Having some control over when to watch the ads is way more user-friendly than having the platform arrange it for you." This feedback affirms the perceived benefits of implementing user-controlled deferral methods for advertisement viewing during video playback. P1 also commented, "When watching videos with this application, I don't have to worry about getting interrupted in the middle of an exciting moment."

Interestingly, all participants expressed a preference for the Ads-debt feature over the Snooze feature. Many of them pointed out that, although snoozing ads videos does delay advertisements, it does not prevent them from reappearing at inopportune moments. This scenario could potentially cause even more interruptions, as it would necessitate additional use of the Snooze feature. The anticipated reappearance of the ads disrupted their viewing experience and made it difficult for them to "fully concentrate on watching the video." (P7) Thus, despite the flexibility offered by Snooze, participants felt that Ads-debt provided a more effective solution for managing the timing of ad interruptions.

Moreover, most participants pointed out that the pop-up panel which prompts users to select the snoozing duration is in itself disruptive. This contradicts the original aim of the feature, which is to minimize interruptions. When asked about potential improvements, P4, for example, suggested that a default or user-set duration could help avoid the need for users to select a snooze duration each time an advertisement is about to play, thereby allowing users to delay the ad without having to make decisions in the middle of a video. In a similar vein, P5 suggested the implementation of a one-click mechanism to reduce the disruption, noting, "It's much more convenient to simply press a button to delay the ads, rather

than having to spend time during the video choosing the snoozing duration."

3.2 Ads Debt Feedback

The feedback on the Ads-debt feature was largely positive. Participants universally agreed that the Ads-debt feature gave them a greater sense of control over the timing of advertisements, which they greatly appreciated. However, some participants did identify potential pitfalls of the feature. For example, they mentioned that when the queue of deferred ads reached its limit, they felt little incentive to clear the queue. Instead, they found it easier to let incoming ads play at their predetermined times, as they *"don't have the time to clear all the ads (P8)."*

P4 suggested an alternative approach where the limitation would be set on a monthly quota instead. In this scenario, if the accumulated ads debt for the current month was cleared, the quota for the next month would increase. This, P4 believed, would motivate users to clear their queue of deferred ads, as it creates the perception of *"receiving rewards for paying back the ads."*

Finally, P9 pointed out that the Ads-debt feature could be particularly effective for users watching live streams. For these viewers, the fear of missing out on crucial moments due to ad interruptions could motivate them to "pay back" their ad debt prior to starting the live stream. These valuable insights offer promising directions for future modifications and enhancements to the Ads-debt feature of the AdJustMoment system.

3.3 Other Findings

The feedback provided by participants underscored the desire for more control over their viewing experience, as well as the value of incentives. Some participants indicated they might choose to perpetually delay or even skip ads, especially when they are primarily interested in the video content itself. For these users, they *"only want to watch the video itself within the limited time"* (P5) and perceive watching ads as a waste of their time.

However, participants also proposed situations where they would be more inclined to watch ads in their entirety. For instance, P2 suggested that ads could provide some kind of enhancement to the video content or even result in rewards, which would make the viewing of ads less of a perceived loss. Similarly, P6 noted that many video platforms require subscriptions to access certain content, and suggested a system where watching ads could *"earn users tokens that could be redeemed"* to unlock premium content. This approach would provide users with a tangible reward and could motivate them to engage more willingly with ads. P3 added another layer of control and interest to the ad-viewing experience by suggesting users could be given a choice of which ad to watch at any given time. This not only provides users with more control over their viewing experience, but it also adds an element of interest, making it *"similar to stumbling upon an interesting short video while watching the original video."*

These findings highlight the importance of user control in the viewing experience and indicate that a reward or incentive system might enhance user engagement with ads. These insights could inform future design improvements for the AdJustMoment system.

4 CONCLUSIONS AND FUTURE WORK

In this study, we have explored solutions to the persistent challenge of interruptions during mobile video streaming due to advertisement intrusions. Drawing inspiration from previous research on deferring notifications, our aim was to determine whether these deferral strategies could also effectively delay ad appearances. To this end, we developed and tested two deferral methods, Snooze and Ads-debt, using a custom-designed application. The qualitative feedback we received indicated that users found these methods superior to current ad delivery methods in mainstream video streaming applications. This was largely because our methods granted users greater control over when ads appeared, reducing the frequency of unwelcome interruptions. Among the two methods, participants displayed a stronger preference for Ads-debt over Snooze, due to the former providing them with a higher degree of control. However, we did encounter potential hurdles related to user willingness to engage with these methods, particularly in the case of Ads-debt. Several participants suggested incorporating rewards to incentivize users to "pay the debt" and watch the ads, which may enhance the appeal of these deferral methods.

Moving forward, we suggest that future designs explore the integration of user control and rewarding mechanisms. This approach could help to strike a balance between minimizing interruptions and meeting the marketing objectives of ads. By designing and testing additional mechanisms, we aim to enhance our understanding of the optimal timing for ad appearances, ultimately improving the overall experience of mobile video streaming.

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